RAM KRISHNA YADAV

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PROFILE

Computer Engineering undergraduate with a background in concurrency and distributed infrastructure, now moving toward machine learning. Author of an open-source JavaScript thread-pool library with over 10,000 npm downloads. Built a retrieval-augmented generation system over vector embeddings and a self-hosted deployment platform for bare-metal servers. Seeking structured training in machine learning theory, mathematical foundations, and research methodology.

EDUCATION

Bachelor of Computer Engineering | 2024 - 2028 (expected)
Tribhuvan University, *Institute of Engineering*, *Purwanchal Campus*, *Dharan*, *Nepal*

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C++

Machine Learning and AI: Retrieval-augmented generation (RAG), vector embeddings, semantic search, pgvector, LLM integration, Model Context Protocol (MCP)

Systems and Concurrency: Worker threads, thread pools, event-driven architecture, WebSockets, Docker, Linux, Nginx, VPS and AWS EC2 deployment

Databases and Storage: PostgreSQL, MySQL, MongoDB, Redis, publish/subscribe messaging, AWS S3

Frameworks and Tools: Node.js, Express.js, Next.js, React, Elysia.js, Drizzle ORM, Git, Jest

OPEN SOURCE

Thready.js - JavaScript Thread Pool Library | npmjs.com/package/thready-js
10,000+ downloads | github.com/imramkrishna/thready-js

- Designed a type-safe thread-pool abstraction exposing one API across Web Workers and Node.js worker_threads.
- Implemented worker pooling, task queueing, and lifecycle management in a zero-dependency architecture.
- Abstracted low-level concurrency primitives into a developer-facing API grounded in JavaScript's event loop and worker execution model.

PROJECTS

EliteNotes - Retrieval-Augmented Generation System | elitenotes.xyz
Next.js, *TypeScript*, *PostgreSQL*, *pgvector*, *Redis*, *Docker*

- Built a multi-format ingestion pipeline (PDF, DOCX, XLSX, CSV) with recursive text chunking and 768-dimensional vector embeddings stored in pgvector.
- Implemented vector similarity search with configurable top-K retrieval and tunable thresholds, producing source-cited, context-grounded responses.
- Designed multi-provider model routing with SSE streaming, token and cost telemetry, and Redis-backed session caching.
- Integrated tool orchestration across 120+ services via the Model Context Protocol (MCP) with dynamic tool loading.

Zyotra - Self-Hosted Deployment Platform (PaaS) | github.com/Zyotra
TypeScript, *Bun*, *Nginx*, *PostgreSQL*, *MySQL*, *Redis*, *WebSockets*, *AWS S3*, *Docker*

- Built a self-hosted platform-as-a-service replicating managed deployment workflows on bare-metal servers, cutting infrastructure cost by up to 80%.
- Engineered agentless deployment automation over SSH, handling repository cloning, TLS certificate provisioning, and reverse proxy configuration.
- Architected real-time build log streaming and live CPU and memory telemetry over WebSockets for deployment observability.
- Implemented automated PostgreSQL and MySQL backups to object storage, single-command Redis provisioning, and an integrated SQL query runner.

ChessOnline - Real-Time Multiplayer System | github.com/imramkrishna/ChessOnline
React, *Node.js*, *Express.js*, *WebSockets*, *MongoDB*, *TypeScript*

- Built a real-time multiplayer system with authoritative server-side state, low-latency WebSocket transport, and complete rule enforcement.
- Engineered an event-driven socket layer handling move broadcast, state synchronization, and graceful disconnect and reconnect recovery.
- Designed the backend to manage concurrent game sessions and real-time player matching.

AREAS OF INTEREST

Machine learning systems and efficiency, inference optimization, distributed training, and NLP for low-resource languages.